

$$1. \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x =$$

$$2. (x^a)' =$$

$$3. (a^x)' =$$

$$4. (e^x)' =$$

$$5. (\log_a x)' =$$

$$6. (\ln |x|)' =$$

$$7. (\sin x)' =$$

$$8. (\cos x)' =$$

$$9. (\tan x)' =$$

$$10. (\cot x)' =$$

$$11. (\sec x)' =$$

$$12. (\csc x)' =$$

$$13. (\ln |\cos x|)' =$$

$$14. (\ln |\sin x|)' =$$

$$15. (\ln |\sec x + \tan x|)' =$$

$$16. (\ln |\csc x - \cot x|)' =$$

$$17. (\ln |\csc x + \cot x|)' =$$

$$18. (\arcsin x)' =$$

$$19. (\arccos x)' =$$

$$20. (\arctan x)' =$$

$$21. (\operatorname{arccot} x)' =$$

$$22. [\ln(x + \sqrt{x^2 + 1})]' =$$

$$23. [\ln(x + \sqrt{x^2 - 1})]' =$$

$$24. \left(\frac{1}{2} \arcsin x + \frac{x}{2} \sqrt{1 - x^2}\right)' =$$

$$25. [\ln(x + \sqrt{x^2 + a^2})]' =$$

$$26. [\ln(x + \sqrt{x^2 - a^2})]' =$$

$$27. (x \ln x)' =$$

$$28. \left(\frac{1}{x} \ln x\right)' =$$

$$29. (e^{ax+b})^{(n)} =$$

$$30. [\sin(ax + b)]^{(n)} =$$

$$31. [\cos(ax + b)]^{(n)} =$$

$$32. [\ln(ax + b)]^{(n)} =$$

$$33. \left(\frac{1}{ax + b}\right)^{(n)} =$$

$$34. \left(\frac{1}{1 + x}\right)^{(n)} =$$

$$35. \int x^k \, dx =$$

$$36. \int \frac{1}{x} \, dx =$$

$$37. \int e^x \, dx =$$

$$38. \int a^x \, dx =$$

$$39. \int \sin x \, dx =$$

$$40. \int \cos x \, dx =$$

$$41. \int \tan x \, dx =$$

$$42. \int \cot x \, dx =$$

$$43. \int \frac{1}{\cos x} \, dx =$$

$$44. \int \frac{1}{\sin x} \, dx =$$

$$45. \int \frac{1}{\cos^2 x} \, dx =$$

$$46. \int \frac{1}{\sin^2 x} \, dx =$$

$$47. \int \sec x \tan x \, dx =$$

$$48. \int \csc x \cot x \, dx =$$

$$49. \int \frac{1}{1 + x^2} \, dx =$$

$$50. \int \frac{1}{a^2 + x^2} \, dx =$$

$$51. \int \frac{1}{\sqrt{1 - x^2}} \, dx =$$

$$52. \int \frac{1}{\sqrt{a^2 - x^2}} \, dx =$$

$$53. \int \sqrt{1 - x^2} \, dx =$$

$$54. \int \sqrt{a^2 - x^2} \, dx =$$

$$55. \int \frac{1}{\sqrt{x^2 + 1}} \, dx =$$

$$56. \int \frac{1}{\sqrt{x^2 + a^2}} \, dx =$$

$$57. \int \frac{1}{\sqrt{x^2 - 1}} \, dx =$$

$$58. \int \frac{1}{\sqrt{x^2 - a^2}} \, dx =$$

$$59. \int \sqrt{1 + x^2} \, dx =$$

$$59. \int \frac{1}{x^2 - a^2} dx =$$

$$61. \int \sin^2 x dx =$$

$$63. \int \tan^2 x dx =$$

$$60. \int \frac{1}{a^2 - x^2} dx =$$

$$62. \int \cos^2 x dx =$$

$$64. \int \cot^2 x dx =$$

$$65. \int_0^{\frac{\pi}{2}} \sin^n x dx =$$

$$69. \int_0^{2\pi} \sin^n x dx =$$

$$66. \int_0^{\frac{\pi}{2}} \cos^n x dx =$$

$$70. \int_0^{2\pi} \cos^n x dx =$$

$$67. \int_0^{\pi} \sin^n x dx =$$

$$71. \int e^{ax} \cos bx dx =$$

$$68. \int_0^{\pi} \cos^n x dx =$$

$$72. \int e^{ax} \sin bx dx =$$

$$1. \sin\left(\frac{\pi}{2} \pm \alpha\right) =$$

$$6. \cos(\alpha \pm \beta) =$$

$$11. \cos 2\alpha =$$

$$2. \sin(\pi \pm \alpha) =$$

$$7. \tan(\alpha \pm \beta) =$$

$$12. \sin 3\alpha =$$

$$3. \cos\left(\frac{\pi}{2} \pm \alpha\right) =$$

$$8. \tan\left(\frac{\pi}{4} - \alpha\right) =$$

$$13. \cos 3\alpha =$$

$$4. \cos(\pi \pm \alpha) =$$

$$9. \cot(\alpha \pm \beta) =$$

$$14. \sin^2 \frac{\alpha}{2} =$$

$$5. \sin(\alpha \pm \beta) =$$

$$10. \sin 2\alpha =$$

$$15. \cos^2 \frac{\alpha}{2} =$$

$$1. e^x =$$

$$2. \sin x =$$

$$3. \cos x =$$

$$4. \arctan x =$$

$$5. \frac{1}{1-x} =$$

$$6. \frac{1}{1+x} =$$

$$7. \ln(1+x) =$$

$$8. (1+x)^a =$$

$$9. \tan x =$$

$$\arcsin x =$$

1. 七种未定式：

2. 等差数列通项公式：

3. 等差数列前 n 项和：

4. 等比数列通项公式：

5. 等比数列前 n 项和：

6. $2f(x) \cdot f'(x) =$

7. $[f'(x)]^2 + f(x)f''(x) =$

8. $[f'(x) + f(x)\varphi'(x)]e^{\varphi(x)} =$

9. $f'(x) + f(x) =$

10. $f'(x) - f(x) =$

11. $f'(x) + kf(x) =$

1. 有界与最值定理：

2. 介值定理：

3. 平均值定理：

4. 零点定理：

5. 费马定理：

6. 导数零点定理：

7. 罗尔定理：

8. 拉格朗日中值定理：

9. 柯西中值定理：

12. 曲率公式：

13. 曲率半径公式：

14. 牛顿-莱布尼茨公式：

15. 区间再现公式：

16. $u''v + 2u'v' + uv'' =$

17. $\frac{xf'(x) - f(x)}{x^2} =$

18. $\frac{f''(x)f(x) - [f'(x)]^2}{f^2(x)} =$

19. $\Gamma(\alpha) =$

1. 若 $a_1, a_2, \dots, a_m \geq 0$, 则 $\lim_{n \rightarrow \infty} \sqrt[n]{a_1^n + a_2^n + \dots + a_m^n} =$

2. 三角不等式

10. $\frac{x}{1+x} \leq \ln(1+x) \leq x, (x > 0)$

3. 反三角不等式

11. $0 < x < \frac{\pi}{2}$

4. 基本不等式

12. $\frac{1}{n^2+n} \leq \frac{1}{n^2+k} \leq \frac{1}{n^2+1}$

5. 三元均值不等式

13. $n \leq \sqrt{n^2+k} \leq \sqrt{n^2+n}$

6. $0 < a < x < b, 0 < c < y < d$

14. $x > 0, \sin x \leq x$

7. $e^x \geq 1+x$

15. $0 < x < \frac{\pi}{4}$

8. $\ln x \leq x-1 (x > 0)$

16. $x > 0, \arctan x \leq x \leq \arcsin x$

9. $\frac{1}{1+x} \leq \ln\left(1+\frac{1}{x}\right) \leq \frac{1}{x}, (x > 0)$